SCIENCE

Textbook for Class X

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विद्यया ऽमृतमश्नुते

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FOREWORD

The National Curriculum Framework, (NCF), 2005, recommends that children's life at school must be linked to their life outside the school. This principle marks a departure from the legacy of bookish learning which continues to shape our system and causes a gap between the school, home and community. The syllabi and textbooks developed on the basis of NCF signify an attempt to implement this basic idea. They also attempt to discourage rote learning and the maintenance of sharp boundaries between different subject areas. We hope these measures will take us significantly further in the direction of a child-centred system of education outlined in the National Policy on Education (1986).

The success of this effort depends on the steps that school principals and teachers will take to encourage children to reflect on their own learning and to pursue imaginative activities and questions. We must recognise that, given space, time and freedom, children generate new knowledge by engaging with the information passed on to them by adults. Treating the prescribed textbook as the sole basis of examination is one of the key reasons why other resources and sites of learning are ignored. Inculcating creativity and initiative is possible if we perceive and treat children as participants in learning, not as receivers of a fixed body of knowledge.

These aims imply considerable change in school routines and mode of functioning. Flexibility in the daily time-table is as necessary as rigour in implementing the annual calendar so that the required number of teaching days are actually devoted to teaching. The methods used for teaching and evaluation will also determine how effective this textbook proves for making children's life at school a happy experience, rather than a source of stress or boredom. Syllabus designers have tried to address the problem of curricular burden by restructuring and reorienting knowledge at different stages with greater consideration for child psychology and the time available for teaching. The textbook attempts to enhance this endeavour by giving higher priority and space to opportunities for contemplation and wondering, discussion in small groups, and activities requiring hands-on experience.

The National Council of Educational Research and Training (NCERT) appreciates the hard work done by the textbook development team responsible for this book. We wish to thank the Chairman of the advisory group in science and mathematics, Professor J.V. Narlikar and the Chief Advisor for this book, Professor Rupamanjari Ghosh, School of Physical Sciences, Jawaharlal Nehru University, New Delhi, for guiding the work of this committee. Several teachers contributed to the development of this textbook; we are grateful to them and their principals for making this possible. We are indebted to the institutions and organisations which have generously permitted us to draw upon their resources, material and personnel. We are especially grateful to the members of

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the National Monitoring Committee, appointed by the Department of Secondary and Higher Education, Ministry of Human Resource Development under the Chairmanship of Professor Mrinal Miri and Professor G.P. Deshpande, for their valuable time and contribution. As an organisation committed to systemic reform and continuous improvement in the quality of its products, NCERT welcomes comments and suggestions which will enable us to undertake further revision and refinement.

Director
National Council of Educational
Research and Training

New Delhi
20 November 2006

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PREFACE

This textbook of Science for Class X is a continuation of our attempt in the Class IX Science textbook to comply with the guidelines of the National Curriculum Framework-2005. We had to work within a limited time frame and also had our own constraints coming in the way of this radical change. The revised and re-structured syllabus for Class X covers selected topics in the broad themes of — Materials, The World of the Living, How Things Work, Natural Phenomena and Natural Resources. We have interpreted the syllabus to present a coherent coverage of scientific concepts related to our daily life on the select topics. It is an integrated approach to science at this level, with no sharp divisions into disciplines such as Physics, Chemistry, Biology and Environmental Science.

There has been a conscious attempt to address the relevant social concerns in this science textbook wherever possible — the concerns for people with special needs, the issues of gender discrimination, energy and environment have found their natural place in this book. Students have been encouraged to get into the debates on some of the management concerns (for sustainable development, for example) so that they can arrive at their own decisions after a scientific analysis of all the facts.

This book has some features which are meant to enhance its effectiveness. The theme of each chapter has been introduced with examples from daily life, and if possible, by a relevant activity that the students have to perform. The entire approach of the book is, in fact, activity-based, i.e., the students are required to construct knowledge themselves from these activities. The emphasis is not on definitions and technical terms, but on the concepts involved. Special care has been taken so that the rigour of science is not lost while simplifying the language. Difficult and challenging ideas, which are not to be covered at this stage, have often been placed as extra material in the boxes in light orange. The excitement of doing science comes from pursuing the unknown — the students would have the opportunity to think and explore somewhat beyond the syllabus and may feel the urge to continue their scientific expedition at higher levels. All such box items, including brief biography of scientists, are, of course, non-evaluative.

Solved examples are provided, wherever felt necessary, to clarify a concept. The in-text questions after a main section are for the students to check their understanding of the topic. At the end of each chapter, there is a quick review of the important points covered in the chapter. We have introduced some multiple choice questions in the exercises. There are problems of different difficulty levels answers to the multiple-choice questions and numericals, and hints for the difficult questions are included at the end of the book.

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This book has been made possible because of the active participation of many people. I wish to thank Professor Krishna Kumar, *Director,* NCERT, Prof. G. Ravindra, *Joint Director,* NCERT, and Professor Hukum Singh, Head, Department of Education in Science and Mathematics, NCERT, specially for their keen interest in the development of the book and for all the administrative support. I wish to put on record my sincere appreciation for Dr Anjni Koul, the member-coordinator of the textbook development committee, for her extraordinary commitment and efficiency. It has been a real pleasure working with my textbook development team and the review committee. The chosen editorial team worked extremely hard, on tight deadlines, to bring the book close to the shape that we dreamt of. Fruitful discussions with some members of the MHRD Monitoring Committee helped in providing the final touches to the book. I do not have the words to acknowledge the professional and personal inputs I received from some of my close friends during the preparation of this book. We warmly welcome comments and suggestions for improvement from our readers.

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THE CONSTITUTION OF INDIA

PREAMBLE

WE, THE PEOPLE OF INDIA, having solemnly resolved to constitute India into a [1]**[SOVEREIGN SOCIALIST SECULAR DEMOCRATIC REPUBLIC]** and to secure to all its citizens :

JUSTICE, social, economic and political;

LIBERTY of thought, expression, belief, faith and worship;

EQUALITY of status and of opportunity; and to promote among them all

FRATERNITY assuring the dignity of the individual and the [2][unity and integrity of the Nation];

IN OUR CONSTITUENT ASSEMBLY this twenty-sixth day of November, 1949 do **HEREBY ADOPT, ENACT AND GIVE TO OURSELVES THIS CONSTITUTION.**

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1. Subs. by the Constitution (Forty-second Amendment) Act, 1976, Sec.2, for "Sovereign Democratic Republic" (w.e.f. 3.1.1977)
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